Stoke's theorem

Let S be a piecewise smooth oriented surface in space and let the boundary of S be a piecewise smooth simple closed curve C. Let $F(x, y, z)$ be a continuous vector function that has continuous first partial derivatives in a domain in space containing S. Then

$$\iint_S \text{curl } F \cdot n \, dA = \oint_C F \cdot r'(s) \, ds.$$

Example

Verify stoke's theorem for $\vec{F} = [y, z, x]$ and S the parabolid

$$z = f(x,y) = 1 - (x^2 + y^2) \quad , \quad z \geq 0$$

Soln:- at $z = 0$ we get

$$x^2 + y^2 = 1$$

which is the equation of circle centered at origin of radius 1. So.

$$\vec{r}(s) = [\cos s, \ \sin s, \ 0]$$

$$\vec{r}'(s) = [-\sin s, \ \cos s, \ 0] \qquad 0 \leq s \leq 2\pi$$

$$F(r(s)) = [\sin s, \ 0, \ \cos s]$$

R.H.S

$$\oint_C F[r(s)] \cdot r'(s)\, ds = \int_0^{2\pi} -\sin^2 s \; ds$$

$$= -\int_0^{2\pi} \frac{(1 - \cos 2s)}{2} ds$$

$$= -\frac{1}{2}\int_0^{2\pi} ds + \frac{1}{2}\int_0^{2\pi} \cos 2s \, ds$$

$$= -\frac{1}{2}(2\pi - 0) + \frac{1}{2}\left(\frac{\sin \ldots}{\ldots}\right)$$

$$= -\pi.$$

15.

L.H.S

$$= \iint_S \text{curl}\, F \cdot n \, dA$$

$$F = [y, z, x]$$

$$\text{curl}\, F = \nabla \times F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix}$$

$$= (-1, -1, -1)$$

Normal vector of S

$$N = \text{grad}(z - f(x,y))$$

$$= \text{grad}(z - 1 + x^2 + y^2)$$

$$= [2x, 2y, 1]$$

$$\text{curl}\, F \cdot N = (-1, -1, -1) \cdot (2x, 2y, 1)$$

$$= -2x - 2y - 1$$

Note $n\, dA = N\, dx\, dy$.

Using polar coordinates r, θ, we define

$x = r\cos\theta$ $\quad 0 \le r \le 1$

$y = r\sin\theta$ $\quad 0 \le \theta \le 2$

$$\iint_R \text{Curl}\,F \cdot N\,dx\,dy = \iint_R (-2x - 2y - 1)\,dx\,dy$$

$$= \int_0^{2\pi}\int_0^1 (-2r\cos\theta - 2r\sin\theta - 1)\cdot r \cdot dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 \left[-2r^2(\cos\theta - \sin\theta) - r\right] dr\,d\theta$$

$$= \int_0^{2\pi} \left[-\frac{2}{3}r^3(\cos\theta - \sin\theta) - \frac{r^2}{2}\right]_0^1 d\theta$$

$$= \int_0^{2\pi} \left[\frac{-2}{3}(\cos\theta - \sin\theta) - \frac{1}{2}\right] d\theta$$

$$= -\frac{2}{3}\left[-\sin\theta - \cos\theta\right]\Big|_0^{2\pi} - \frac{1}{2}\theta\Big|_0^{2\pi}$$

$$= +\frac{2}{3}\left(\cancelto{0}{\sin 2\pi} - \cancelto{0}{\sin 0}\right) + \frac{2}{3}(\cos 2\pi - \cos 0)$$

$$- \frac{1}{2}(2\pi - 0)$$

$$= 0 + \frac{2}{3}(1-1) - \pi = -\pi$$